Multimodal, Interactive Systems, UXD-M
Exercise, Summer term 2027

Exercise Sheet 05
Prof. Dr. Simon Nestler

Audit and one implemented substitution

What is this sheet about?

This sheet audits your own prototype for barriers – in two columns, because an automated checker finds only a fraction of them and the rest need a human. You then build one modality substitution that actually removes a documented barrier, and you write down what it costs other users. A substitution is never free, and naming the side effect is part of the work.

Docking point: This exercise cashes in sessions T10 (*Reported green, sued anyway*) and T11 (*The ramp that gets in the way*): you know the checker finds 57, 40, or 14 percent of the barriers depending on whom you believe, and you know a modality substitution is never free ; today both lessons apply to your own prototype.

By the end of this block your team has the milestone from the semester plan: an **audit report with one implemented improvement**. Concretely: a two-column audit of your exercise-block-4 prototype (what the checker found versus what only a human found), plus one modality substitution, actually built into the prototype, that removes one documented barrier, with its side effect for other users named in writing.

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Organisation: Please work on this exercise sheet in your project team (fixed since exercise block 1). The first and second assignment are done in pairs within the team (rotating for the manual pass); the team members are called A, B, C (and D, if you are working in a team of four). In order to avoid confusions please document at the beginning who will take which (generic) role.

AI role: tool and object of examination (both declared). Tool, because you run automated checkers; object of examination, because you audit the checker's findings against your own manual pass. Trusting the green light is exactly the mistake T10 was about. Your upload **must disclose what was made with which tool**.

Bonus points: In order for the bonus rule to apply (participation in all six blocks with an upload within 48 hours each earns 5 % compensation on the exam), your team must hand in one complete submission package (see the end of this sheet) within 48 hours after this exercise. This is block five of six.

Tools: *WAVE* or *axe* for the automated pass, both known from the T09→T10 bridge task; they apply to the web- and screen-based parts of your prototype. Where your prototype has no auditable web surface (voice-only, gesture-only), the checker's silence is itself a finding: note what the tool cannot even see. *VoiceOver* (iOS/macOS) or *TalkBack* (Android) for the screenreader pass, or *Narrator* (Windows logo key + Ctrl + Enter) if your prototype runs on a Windows laptop. Learn the off switch before you switch the screen off. You have 15 minutes of self-experiment from the T10 bridge task; today you sit on the other side of it. You also need your own prototype from exercise block 4, running. **If it does not run, the first 30 minutes of this block are repair time and your audit shrinks.** The rule is announced here so that your team can plan for it.

First assignment (25 minutes): Automated pass, in pairs within the team. Run WAVE or axe over every auditable surface of your prototype. Record: error count, warnings, and — T10 discipline — the findings you do not understand, verbatim. Separately, list the parts of your prototype the checker could not reach at all (the gesture channel, the voice flow). That list is column-one material for the report. If your prototype opens with a consent or permission dialog, audit that dialog first: it is the one screen every user must pass.

Second assignment (40 minutes): Manual pass, in pairs, rotating. Three sub-assignments, mirroring the T10 live test, plus one optional:

- **a)** Operate the full task with keyboard or single-switch only: **no mouse, no touch**, and no pointer emulation (*Pointer Control* on macOS or *Mouse Keys* on Windows turn the keyboard back into a mouse and hide the barrier you are looking for). Check at least WCAG 2.2 SC 2.1.1 Keyboard, 2.1.2 No Keyboard Trap, 2.1.4 Character Key Shortcuts, and 2.4.7 Focus Visible.
- **b)** Screenreader on, **screen off or averted**, full task.
- **c)** One person describes every image, icon, and state signal aloud to a partner who cannot see the screen. What cannot be described was never designed.
- **d)** *(optional, only if the 40 minutes allow)* Operate three steps by voice only with *Voice Control* (iOS/macOS) or *Voice Access* (Android): speak the visible label of each control. Every control you cannot name aloud has a label problem.

Protocol each break-off point: where, why, which modality assumed which ability.

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Third assignment (20 minutes): Two-list confrontation, as a team. Merge the passes: what did only the checker find, what did only the human find, what did both find? For each human-only finding, one sentence on *why* the automated pass missed it. Tag every human-only finding with the information-presentation principle it violates (DIN EN ISO 9241-112): *detectability*, *freedom from distraction*, *discriminability*, *unambiguous interpretability*, *conciseness*, or *consistency*. These principles hold for sound, speech, and vibration as well as for screens. This table is your own small replication of the three-numbers controversy, with numbers you collected yourselves.

Fourth assignment (45 minutes): Design and implement one modality substitution, as a team. Pick one barrier from the human-found list that a real user would hit, even if an easier one is on the list. Design the substitution or addition that removes it (audio→text, visual→audio, touch→speech, or the reverse), name **who gains and who pays** — the T11 rule that substitution is never free — and build it into the prototype. Test your gains-and-pays claim against four people:

- a woman of 40 who lost her sight a few weeks ago through an illness and still relies on an assistant;
- a 19-year-old with an intellectual disability who wants to manage on his own, with his parents reachable by phone;
- a man of 70 with 10–20 % of his vision left, supported by a household help;
- a deaf father of 35 who does not speak and is often, but not always, accompanied by his hearing 15-year-old daughter, who signs.

Rerun the manual sub-assignment that found the barrier and record whether it now passes.

Fifth assignment (20 minutes): Write the audit report, as a team. Assemble findings, table, substitution, side effect, and retest result into the report. Write it for a reader who was not in the room; your exercise-block-6 test participants and the exam-preparing version of yourself both qualify.

Buffer/critique (30 minutes): Silent gallery: audit tables go on the wall or shared screen; each team reads two neighbors' tables and leaves one written question each. Answer the questions you received in one sentence in the report — or mark them honestly as open. The buffer also absorbs screenreader friction, which for first-time auditors is guaranteed.

Submission (within 48 h, one upload per team):

- `audit-report.md` — the full report: automated findings, manual findings, the two-list table with per-finding explanations, the substitution with named side effect, retest result, and the two neighbor questions with answers (max 5 pages).
- `prototype/` — the updated prototype containing the implemented substitution, with `run.md`.
- Before/after evidence of the removed barrier (2 recordings, max 60 seconds each, .mp4, max 100 MB total).
- **AI disclosure line** (in `audit-report.md`, mandatory): *"Made with AI: [tool] was used for [part]; checked and understood by [name]."*

Put everything into one upload per team and hand it in via Moodle within 48 hours after this exercise.